MILITARY SPECIFICATION

BATTERY, WET PRIMARY

FOR AN/DKT-37(V) TELEMETRY PACKAGE

"This specification is approved for use by Naval Air Systems Command Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the performance, design, test, manufacture, and acceptance requirements for the wet primary battery critical item, referred to herein as the "battery".

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, from a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated - Plywood

Military

MIL-D-1000	Drawings and drawing practices
MIL-Q-9858	Quality Assurance Requirements
MIL-C-45662	Calibration System Requirements

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage

Comments or recommendations which may be of use in improving this document may be sent to the preparing activity when a Self-Addressed Standardization Document Improvement Proposal (DD Form 1426) appears at the end of this document, or by letter adddessed to the cognizant activity.

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MIL-STD-130	Identification for Marking of US Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Components Parts
MIL-STD-210B	Climatic Extremes for Military Equipment
MIL-STD-454	Standard General Requirement for Electronic Equipment
MIL-STD-480	Configuration Control-Engineering Changes, Deviations and Waivers
MIL-STD-481	Configuration Control-Engineering Changes, Deviations and Waivers
MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, preparation of
MIL-STD-1285	Marking of Electrical and Electronic Parts

DRAWINGS

Whittaker Corporation
(Code Ident 16405)

203655	Battery, Wet Primary

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS.

3.1 Item Definition. The battery is a wet primary, remotely activated, hermetically sealed, silver oxide-zinc battery.

3.2 Characteristics. The electrical requirements specified herein shall be met under any combination of environmental conditions specified herein over the temperature range of -50 to +160 degrees Fahrenheit (^{o}F).

3.2.1 Performance. The following battery performance requirements shall be met when subjected to the test of section 4 of this specification.

3.2.1.1 Output Voltage. The battery output voltage, with the load as specified in Figure 1 and tested in accordance with 4.5.5, shall be from 24.0 to 32.0 volts

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direct current (vdc).

3.2.1.1.1 Activation Time. The battery output voltage shall not be less than 24.0 vdc minimum after 0.5 second from application of igniter initiation current.

3.2.1.2 Leakage Current. Leakage current between the positive terminals and case shall not exceed 0.1 ampere (A) direct current (dc).

3.2.1.3 Noise Level. The noise level of the battery output voltage while under steady state load shall not exceed 150 millivolts (MV) peak-to-peak (p-p) from 0 to 5 kilohertz (kHz) when tested as specified in 4.5.5 and 4.5.6.

3.2.1.4 Active Life. The battery output voltage shall be within the limits specified for 90 seconds minimum after activation.

3.2.1.5 Heaters.

3.2.1.5.1 Heater Power. The battery shall be equipped with thermostatically controlled electric heater(s). The heater shall operate from 113.5 + 6 volts root-mean-square (VRMS) at 400+20 Hertz (Hz), single phase, as defined by MIL-STD-704, Category B. The current shall not exceed 2.5 A.

3.2.1.5.2 Heater Warm-Up Period. The warm-up period shall be 10 minutes+15 seconds. The battery shall meet the requirements of 3.2.1 after the completion of the warm-up period, in a still-air environment at any temperature within the range of -50° to $+160^{\circ}$F.

3.2.1.6 Igniter.

3.2.1.6.1 Igniter All-Fire Current. The battery shall activate within 10 milliseconds (ms) when a direct current of 1.6 + 0.1 A is applied to the igniter activation terminals for not more than 30 ms. (See drawing 203655).

3.2.1.6.2 Igniter No-Fire Current. The battery shall not activate when a direct current of 0.096 to 0.116 A is applied to the igniter activation terminals (See drawing 203655).

3.2.1.6.3 Igniter Circuit Resistance. The battery igniter (squib) resistance across activator contacts 1 and 6 shall be 3.5 $\pm$ 1.0 ohm (Ω).

3.2.1.7 Insulation Resistance. The battery insulation resistance shall be as shown in Table I.

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Table I. Insulation Resistance

Pin Connector From	Pin Connector To	Insulation Resistance (MΩ) Minimum
1, 6	Case	50
1, 6	2, 3	10
1, 6	4, 5	10
1, 6	7, 9	10
2, 3	Case	50
2, 3	7, 9	10
4, 5	Case	50
4, 5	7, 9	10
7, 9	Case	50

Megohms (MΩ)

3.2.2 Physical Characteristics.

3.2.2.1 Weight. The total weight of the battery shall not exceed 2.2 pounds.

3.2.2.2 Physical Dimensions. The battery physical dimensions shall be in accordance with Drawing 203655.

3.2.2.3 Seal. The battery shall be a hermetically sealed unit and shall show no evidence of leakage when tested in accordance with 4.4.2.

3.2.2.4 Radiographic Inspection. Radiographic inspection shall reveal no assembly or construction discrepancies.

3.2.3 Reliability. The battery shall demonstrate a reliability of 0.995 at a 90 percent confidence level when activated during the environmental conditions specified in 3.2.4. Demonstrations of this requirement shall be through cumulative tabulation of performance data in accordance with reliability data requirements DID-UDI-R-21136.

3.2.4 Environmental Conditions. The battery shall show no detrimental effects when subjected to continuous or intermittent exposure to any single condition or combination of the environmental requirements specified herein:

(a) Low temperature exposure (see 4.6.1.3)

(b) Shock handling (see 4.6.5)

(c) Vibration (see 4.6.4)

(d) Heater cycling (see 4.6.1.3 (d))

(e) Acceleration (see 4.6.6)

(f) Altitude (see 4.6.2)

(g) Humidity (see 4.6.3)

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(h) Low temperature operation (see 4.6.1.2)

(i) High temperature operation (see 4.6.1.2)

3.3 Design and Construction. The design, construction, materials, and finish of the battery shall conform to the requirements specified herein and on Drawing 203655. Where not specified, the details of design, construction, materials, and finish shall be such that the battery will meet all requirements specified herein. The battery shall be designed to have a useful life of five years minimum from date of acceptance.

3.3.1 Materials, Processes, and Parts.

3.3.1.1 Materials. Materials used in the external construction of the battery shall be in accordance with MIL-STD-454, Requirement 4, Fungus-Inert Materials, Requirement 15, Ferrous Alloys, Corrosion Resistance, and Requirement 16, Dissimilar Metals.

3.3.1.2 Welding. Structural welding shall be in accordance with MIL-STD-454, Requirement 13, except radiographic, magnetic particle or liquid penetrant inspections are not required.

3.3.1.3 Parts.

3.3.1.3.1 Battery Connector. The number of contacts, contact arrangement and contact electrical connections shall be in accordance with Drawing 203655. The connector shall be hermetically sealed.

3.3.1.3.2 Battery Case. The case shall be of construction and materials that shall withstand battery operation. During useful life, the battery shall not rupture or deform in a manner that would result in electrolyte leakage.

3.3.1.3.3 Gas Generator. The gas generator used to transfer the electrolyte from storage to the cell block shall be provided with dual electrical igniter bridges in parallel, terminating in four individual leads external to the gas generator case. Each igniter bridge shall be capable of activating the gas generator independent of the second ignitier bridge.

3.3.1.3.4 Temperature Control. A temperature control shall be used for heater operation. An over-temperature safety device shall be incorporated in series with the control.

3.3.2 Identification and Marking. Identification and marking of the battery shall conform to MIL-STD-130 (use of decals is not permitted) and shall include the following:

(a) Manufacturer's name or symbol

(b) Serial number, traceable to all recorded data

(c) Drawing number

(d) Manufacture date, coded in accordance with MIL-STD-1285

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(e) Warning, marked in 0.06/0.10 inch high characters (see Drawing 203655)

(f) Device orientation identification shall be marked in contrasting color approximately where shown (see Drawing 203655)

3.3.3 Workmanship. The battery, including all parts and accessories, shall be constructed and finished in a manner that will assure compliance with all requirements specified herein and shall meet the requirements of MIL-STD-454, Requirement 9.

3.3.4 Interchangeability. The battery shall be interchangeable in accordance with MIL-STD-454, Requirement 7.

3.3.5 Safety. The assembly, loading, and handling of the explosive subassemblies and finished items covered by this specification involve hazardous operations and therefore require suitable explosives safety precautions.

3.3.6 Standards of Manufacture. As a minimum, the battery shall be manufactured under conditions and procedures meeting the requirements specified herein.

3.4 First Article.

3.4.1 Preproduction Sample. Unless otherwise specified in the contract or purchase order, a preproduction sample of 17 batteries shall be delivered to the testing activity designated in the contract or purchase order for preproduction testing. The preproduction sample shall be manufactured using the same design, processes, materials, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor.

3.4.2 Storage Life Sample. When specified in the contract or purchase order, ten batteries shall be delivered to the activity designated in the contract or purchase order for storage life testing in accordance with 4.9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 General. Tests, examinations, and inspections shall be performed on the battery to verify that the requirements of Section 3 have been met.

4.1.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections

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are deemed neccessary to assure that supplies and services conform to prescribed requirements.

4.1.1.1 Production Batteries. Batteries furnished under this specification shall be representative of the product which has been tested and has passed the preproduction inspections specified herein. Before any substitutions are incorporated, approval in writing shall be obtained from the procuring activity in accordance with MIL-STD-481. Samples deemed necessary to verify the suitability of the proposed substitution may be requested at the discretion of the procuring activity.

4.1.1.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.1.2.1.1)

(b) Quality conformance inspection (see 4.2)

4.1.2 Special Tests and Examinations.

4.1.2.1 First Article.

4.1.2.1.1 Preproduction Inspections. The preproduction sample shall be subjected to the tests specified in Table II in the sequence shown and data provided in accordance with DID DI-T-5329.

4.1.2.1.2 Acceptance Criteria. Any battery failing to pass the examinations and tests specified in Table II shall be cause for refusal to grant first article approval. Data requirements must be provided in accordance with DID UDI-R-21141

Table II. Preproduction Tests and Inspections

Examination or Test	Requirement Paragraph	Method Paragraph	Sample Units to be Tested
Group I			
Visual and Mechanical	3.2.2.1, 3.2.2.2, 3.3, 3.3.3 and 3.3.4	4.4.1	
Radiographic	3.2.2.4	4.7	All units
Seal	3.2.2.3	4.4.2	
Igniter no-fire current	3.2.1.6.2	4.5.1	
Igniter circuit resistance	3.2.1.6.3	4.5.2	
Insulation resistance	3.2.1.7	4.5.3	

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Table II. Preproduction Tests and Inspections (Cont.)

Examination or Test	Requirement Paragraph	Method Paragraph	Sample Units to be Tested
Group II			
Low temperature exposure	3.2.4 (a)	4.6.1.3	All Units
Heater cycling	3.2.4 (d)	4.6.1.3	
Igniter circuit resistance	3.2.1.6.3	4.5.2	
Insulation resistance	3.2.1.7	4.5.3	
Group III			
Shock-handling	3.2.4 (b)	4.6.5	All Units
Seal	3.2.2.3	4.4.2	
Igniter circuit resistance	3.2.1.6.3	4.5.2	
Insulation resistance	3.2.1.7	4.5.3	
Activate battery #1 at completion of test	3.2.1	4.5.5 4.5.6.& 4.5.7	Unit 1
Vibration	3.2.4 (c)	4.6.4	Units 2 - 17
Seal	3.2.2.3	4.4.2	
Igniter circuit resistance	3.2.1.6.3	4.5.2	
Insulation resistance	3.2.1.7	4.5.3	Units 2, 3, & 4
Activate batteries 2, 3, & 4 during test	3.2.1	4.5.5, 4.5.6 & 4.5.7	
Acceleration	3.2.4 (e)	4.6.6	Units 5 - 17
Seal	3.2.2.3	4.4.2	
Igniter circuit resistance	3.2.1.6.3	4.5.2	
Insulation resistance	3.2.1.7	4.5.3	Units 5 & 6
Activate batteries 5 & 6 during test	3.2.1	4.5.5, 4.5.6 & 4.5.7	
Group IV			
Altitude	3.2.4 (f)	4.6.2	
Seal	3.2.2.3	4.4.2	Unit 7 - 17

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Table II. Preproduction Tests and Inspections (Cont.)

Examination or Test	Requirement Paragraph	Method Paragraph	Sample Units to be Tested
Group IV (Cont.)			
Igniter circuit resistance	3.2.1.6.3	4.5.2	Unit 7 - 17
Insulation resistance	3.2.1.7	4.5.3	
Activate battery 7 during test	3.2.1	4.5.5, 4.5.6 & 4.5.7	Unit 7
Humidity	3.2.4 (g)	4.6.3	
Seal	3.2.2.3	4.4.2	
Igniter circuit resistance	3.2.1.6.3	4.5.2	
Insulation resistance	3.2.1.7	4.5.3	Unit 8 (only)
Activate battery 8 at completion of test	3.2.1	4.5.5, 4.5.6 & 4.5.7	
Group V			
Low temperature operation	3.2.4 (h)	4.6.1.2	Units 9 - 17
Activate batteries 9, 10, 11, & 12 during test	3.2.1	4.5.5, 4.5.6 & 4.5.7	Units 9, 10, 11 & 12
High temperature operation	3.2.4 (i)	4.6.1.2	
Activate batteries 13,14 15, 16 & 17 during test	3.2.1	4.5.5, 4.5.6 & 4.5.7	Units 13 - 17

4.2 Quality Conformance Inspections. Quality conformance inspections shall consist of Group A and Group B inspections as specified herein.

4.2.1 Group A Inspection. Group A inspection shall consist of the examinations and tests specified in Table III. All batteries shall be subjected to the inspections and tests of Group A and data provided in accordance with applicable DID DI-T-5329.

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Table III. Group A Inspection

Examination or Test	Requirement Paragraph	Method Paragraph
Subgroup I		
Visual and mechanical examination	3.2.2.1, 3.2.2.2, 3.3, 3.3.3, & 3.3.4	4.4.1
Seal	3.2.2.3	4.4.2
Radiographic	3.2.2.4	4.7
Subgroup II		
Igniter no-fire current	3.2.1.6.2	4.5.1, 4.2.1.1 (Step 1 only)
Igniter circuit resistance	3.2.1.6.3	4.5.2, 4.2.1.1
Insulation resistance	3.21.7	4.5.3, 4.2.1.1

4.2.1.1 Subgroup II Test Conditions. Subgroup II tests shall be performed as follows:

Step 1: At room ambient with no heater power applied, the test of Subgroup II.

Step 2: At the completion of Step 1, apply heater power. When heater power has been applied for 10 minutes, then removed, repeat the test of Subgroup II.

Step 3: Condition battery at $-50^{\circ} + 5^{\circ}$F for one hour with no heater power applied. Apply heater power for three minutes, remove power, and repeat the test of Subgroup II.

Step 4: At room ambient with heater power applied for ten minutes, then removed, repeat the test of Subgroup II.

Record the heater "on" time for each operation and the number of times that thermostats open and close in accordance with 6.2.1.

4.2.1.2 Rejected Samples from Group A Tests. Individual batteries rejected as a result of failure to pass Group A testing shall be removed from the lot. A description of all failures in accordance with 6.2.1 shall be supplied with Group A test data. All failures in Group A tests must be analyzed and proposed corrective action submitted to the procuring activity for approval prior to rework. Failures data must be provided in accordance with applicable DID UDI-R-21141.

4.2.2. Group B Inspection. Group B inspection shall consist of the examinations and tests specified in Table IV in the sequence shown. Data must be provided in accordance with applicable DID DI-T-5329.

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Table IV. Group B Inspection

Examination or Test	Requirement Paragraph	Method Paragraph
Shock handling	3.2.4 (b)	4.6.5
Random vibration	3.2.4 (c)	4.6.4
Heater cycle	3.2.4 (d)	4.6.1.3 (d)
Seal	3.2.2.3	4.4.2
Igniter circuit resistance	3.2.1.6.3	4.5.2
Insulation resistance	3.2.1.7	4.5.3
Low temperature operation	3.2.4 (h)	4.6.1.2*
High temperature operation	3.2.4. (i)	4.6.1.2*

*NOTE 1. Number samples in accordance with 6.2.1 beginning with 1 as the first sample selected, and continue sequentially. Subject all odd numbered samples to the high temperature tests. Subject all even numbered samples to the low temperature tests.

4.2.2.1 Lot Formation. For purposes of Group B inspections a lot shall be as defined in MIL-STD-105.

4.2.2.2 Sample Selection. Samples for Group B inspection shall be randomly selected from batteries which have been subjected to and passed the Group A inspections of Table III. The Group B sample size shall be in accordance with Table V.

Table V. Group B Test Samples

Lot Size	Group B Test Samples
1 through 60	14 percent
61 through 150	12 percent
151 and up	10 percent

4.2.2.3 Acceptance Criteria for Group B Inspection. No failures are allowed during Group B tests. In the event a failure occurs, the lot shall be rejected and the procuring activity shall be notified within 24 hours of the failure. Further manufacture of batteries shall be suspended. Failure analysis shall be developed and evaluated to verify effectiveness. A report of analysis results and proposed corrective action with details of evaluation results in accordance with 6.2.1 is required. Manufacture of batteries shall not be resumed until approved by the procuring activity. Data results must be in accordance with applicable DID UDI-R-21141.

4.3 Test Conditions and Provisions.

4.3.1 Test Equipment

4.3.1.1 Accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy of at least one-fifth of the tolerance or 1 percent of the variable to be measured.

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4.3.1.2 Test Equipment Maintenance and Calibration. Unless otherwise specified herein, all test equipment shall be calibrated and maintained in accordance with the requirements of MIL-C-45662.

4.3.1.3 Test Conditions. Unless otherwise specified herein or on the applicable drawing, all test measurements shall be conducted under the following environmental conditions:

Altitude: Elevation of test facility

Temperature: Between 65° and 95°F, ambient

Humidity: 90 percent maximum

4.3.1.4 Heater Power. Heater power of 113.5+6 Vrms at 60+3 Hz may be used during the performance of the tests of 4.1.2.T.1 and 4.2 when the application of heater power is specified. The current shall not exceed 2.5 amps.

4.4 Physical Test and Inspection Methods. Batteries shall meet requirements specified in Tables II, III, and IV as appropriate.

4.4.1 Visual and Mechanical Examinations. The battery shall be visually and mechanically examined for conformance with 3.2.2.1, 3.2.2.2, 3.3, 3.3.3 & 3.3.4.

4.4.2 Seal. The battery shall be subjected to the seal test in accordance with MIL-STD-202, Method 112, Test Condition B, except water shall be used instead of silicone oil.

4.5 Performance Test Methods.

4.5.1 Igniter No-Fire Current. A current of 0.096 to 0.116 Adc shall be applied to the igniter activation terminals (see Drawing 203655) for a period of 30±5 sec.

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4.5.2 Igniter Circuit Resistance. The circuit resistance across the igniter activation terminals (see Drawing 203655) shall be measured with a maximum current of 10 MA.

4.5.3 Insulation Resistance. The insulation resistance shall be measured in accordance with MIL-STD-202, Method 302, Test Condition B.

4.5.4 Activation Current Test. A current of 1.7 + 0.1 Adc shall be applied to the igniter activation terminals (see Drawing 203655) for 30 ms maximum.

4.5.5 Battery Output Voltage Test. When subjected to the load conditions of Figure 1, the battery shall be activated as specified in 4.5.4. The battery output shall meet the requirements of 3.2.1.1 through 3.2.1.4.

4.5.6. Noise Level Test. The noise level shall be measured across the positive and negative terminals through a 0.1 microfared capacitor.

4.6 Service Conditions. The equipment shall operate satisfactorily under any of the following service conditions or natural combinations thereof which are normally encountered during its service life.

4.6.1 Temperature.

4.6.1.1 Storage Temperature. The equipment shall be capable of meeting performance requirements specified herein after storage for 5 years under atmospheric temperature extremes ranging from -65°F to +160°F.

4.6.1.2 Operating Temperature. The equipment shall meet performance requirements specified herein during exposure to maximum adiabatic skin temperatures of -50° to +200°F during captive flight and to free flight temperatures of up to 370°F.

4.6.1.3 Low Temperature Exposure. The battery shall be subjected to the conditions as follows:

(a) Place the battery in a temperature chamber that is stabilized at ambient conditions.

(b) Decrease the temperature within the chamber to -65° $\pm$5°F within a period of 5 hours $\pm$15 minutes.

(c) Maintain the -65° $\pm$5°F temperature for a minimum duration of 72 hours.

(d) Increase the temperature in the chamber to -50° +5°F within 3 hours Allow to stabilize for 1 hour, then apply heater power in accordance with 3.2.1.5. Measure and record heater current number of cyclings, and duration of each cycle for 1 hour in accordance with 6.2.1

(e) Remove batteries from chamber and allow to stabilize for 1 hour at room temperature before resuming testing.

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4.6.2 Altitude. The equipment shall be capable of satisfactory operation from sea level to 90,000 feet altitude (29.92 to 0.520 inches of mercury).

4.6.3 Humidity. The equipment shall be capable of meeting performance requirements specified herein after exposure to relative humiditiy of 95 percent at temperatures up to 95°F during unpacked storage.

4.6.4 Vibration.

4.6.4.1 Flight Vibration. The equipment shall meet performance requirements specified herein while being subjected to input random vibration levels and bandwidths as defined by Figure 2. The equipment shall meet these requirements at ambient; -50°F and +160°F.

4.6.4.2 Captive Flight Vibration. The equipment shall be capable of meeting performance requirements specified herein after exposure to random vibration at a level up to 0.04 g^2/Hz over a frequency range of 50 to 2000 Hz during captive flight. The equipment shall be capable of meeting performance requirements specified herein after 100 captive flights.

4.6.4.3 Transportation Vibration. The equipment shall be capable of meeting performance requirements specified herein after exposure to transportation vibrations of 2 g's over the frequency range of 2 to 2000 Hz for a maximum of 30 days.

4.6.5 Shock.

4.6.5.1 Arrested Landing. The equipment shall be capable of withstanding the shock loads of 100 arrested landings and, thereafter, be capable of meeting the performance requirements.

4.6.5.2 Catapult Launch. The equipment shall be capable of withstanding the shock loads of 100 catapult launches and, thereafter, be capable of meeting the performance requirements.

4.6.5.3 Handling Shock. The equipment shall be capable of withstanding a total of 60 shocks of 35 g's at 11 ms duration in the unpacked condition. The equipment shall meet the performance requirements specified herein after exposure to this condition.

4.6.5.4 Ejection Shock. The equipment shall, while positioned in any plane normal to the longitudinal axis, be capable of meeting the performance requirements during and after ejection launch.

4.6.5.5 Ignition Shock. The equipment shall be capable of satisfactory operation during and after a shock of 50 g's peak, 35g/ms onset and a time duration of not less than 5 milliseconds applied along the longitudinal axis.

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4.6.6 Acceleration. The equipment shall operate satisfactorily during sustained accelerations of not less than 40 g's in the forward direction and 15 g's in the aft direction along the longitudinal axis, +25 g's along any one wing axis (perpendicular to longitudinal axis) and +35 g's on any axis perpendicular to longitudinal axis and rotated 45 degrees from the wing axis.

4.6.7 Salt Fog. The equipment shall be capable of meeting the performance requirements specified herein after exposure to a salt fog environment during unpacked storage.

4.6.8 Sand and Dust. The equipment shall be capable of meeting the performance requirements specified herein after exposure to sand and dust as defined in MIL-STD-210.

4.6.9 Rain. The equipment shall be capable of meeting the performance requirements specified herein after exposure to rainfall as defined in MIL-STD-210.

4.6.10 Acoustic Noise. The equipment shall be capable of meeting the performance requirements specified herein while being subjected to acoustical noise vibrations at a test overall sound pressure level of 160dB (0 dB=2 x 10^{-4} dynes/cm^2).

4.7 Radiographic Inspection. The battery shall be radiographically inspected in accordance with MIL-STD-202, Method 109.

4.8 Preservation, Packaging, Packing, and Package Marking. Prior to shipment, the battery shall be examined to ensure that preservation, packaging, packing, and package marking for shipment are in accordance with the requirements of Section 5.

4.9 Storage Life Testing. Beginning 3 years after the date of manufacture and for each succeeding year thereafter, two batteries shall be withdrawn from storage by the Government and shall be subjected to the Group B Inspection (see Table IV). Upon completion of the above requirement, the batteries shall be activated in accordance with 4.5.5, 4.5.6, and 4.5.7. Data shall be recorded for the requirements of 3.2.1.1, 3.2.1.2, 3.2.1.3, and 3.2.1.4 in accordance with 6.2.1.

5. Preparation for Delivery.

5.1 Preservation and Packaging.

5.1.1 Level C. The unit shall be packed in such a manner as to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. In additon, the packaging shall have a moisture indicator.

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5.2 Packing.

5.2.1 Level C. The unit shall be packed in such a manner as to afford individual protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking. In addition to any marking required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. Notes.

6.1 Intended Use. The battery is intended for use as an electrical power supply during flight in a guided missile.

6.2 Ordering Data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Testing activity designated to test preproduction sample (see 3.4.1)

(c) Storage life sample (see 3.4.2).

(d) Invocation of DOD 4145.26M, DOD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material.

(e) Test equipment accuracies if other than specified.

(f) Preservation, packaging, packing, and applicable marking, if different from Section 5.

6.2.1 Data Requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraphs	Data Requirement	Applicable DID
4.1.2.1.1, 4.2.1, 4.2.2	Test reports	DI-T-5329
3.2.3	Reliability data requirements	UDI-R-21136

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Paragraphs	Data Requirement	Applicable DID
4.2.2.3. 4.1.2.1.2, 4.2.1.2	Failure analysis and report	UDI-R-21141

6.3 Definitions.

6.3.1 Operating Conditions. The operating condition is defined as conformance to the battery output voltage requirements specified in 3.2.1.1.

Preparing activity:
Navy - AS

(Project 6135-N148)

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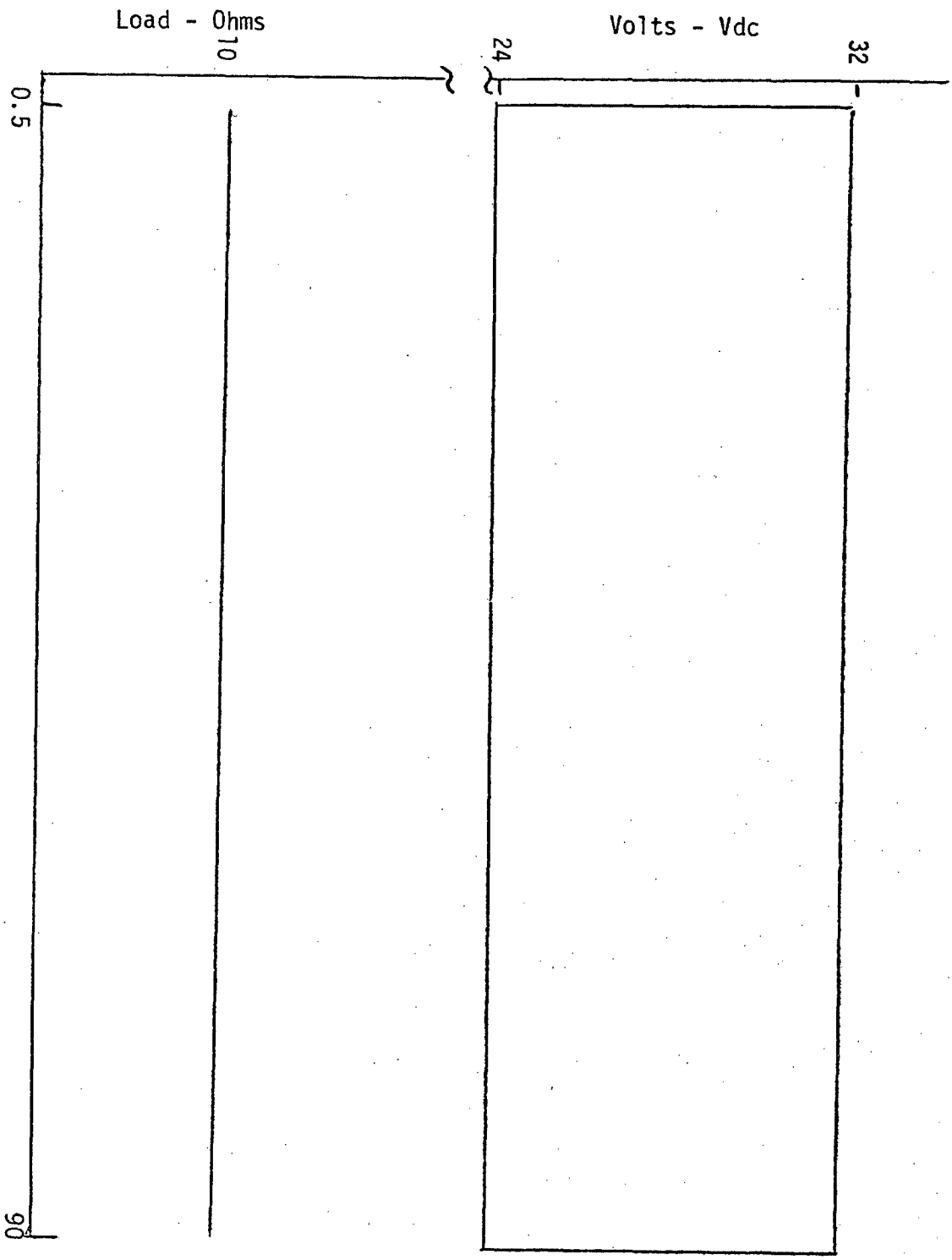
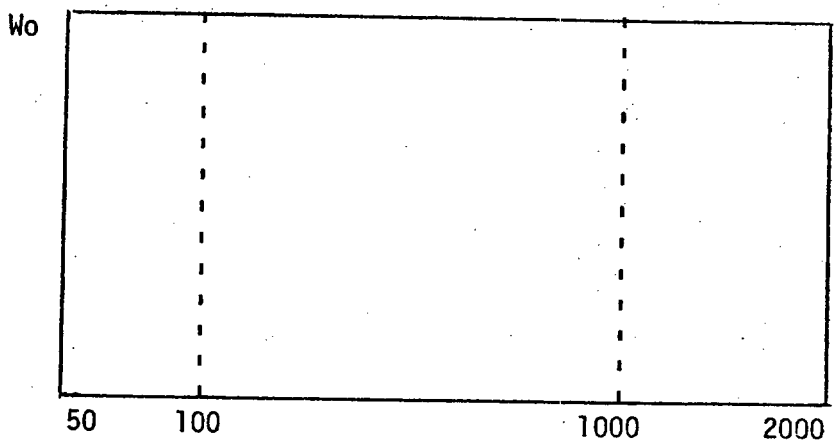


Figure 1. Load Conditions

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Acceleration Power Spectral
Density g^2/cps



NOTES:

1. The value of Wo is 0.04 g^2/Hz, for captive flight and 0.05g^2/Hz for free flight.
2. The overall RMS G minimum is 9.3.
3. The accuracy of the amplitude control shall be as follows:

Frequency Range	Tolerance
0 to 1000 cps	+3.0, -1.5 dB
1000 to 2000 cps	± 3.0 dB

Analysis equipment as described in MIL-STD-810 shall be used for verification of these accuracies.

Figure 2 Random Vibration Test Curve Envelope

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-B-85099(AS) Military Specification Battery, Wet Primary for AN/DKT-37(V) Telemetry Package

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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